NALLABATI VENKATA DURGA SAI SARAT CHANDRA

Target Role: Unmanned Aerial Vehicle (UAV) Autonomy Intern • Skylark Drones

+91 6305761091 • saratchandranallabati@gmail.com • Portfolio: sarat-2007.github.io • GitHub: github.com/sarat-2007 • LinkedIn

Education & National Merit Pedigree

Amrita Vishwa Vidyapeetham — **August 2024 – Expected May 2028**

B.Tech in Robotics & Artificial Intelligence (Cumulative GPA: 9.05 / 10.00, Top 5% Cohort) — *Amritapuri, India*

- **Academic Standing: AP POLYCET State Merit Rank: 497** (Top 0.5% Statewide Percentile).
- **National Distinctions: ISRO-IIRS YUVIKA** (Selected for space science analytics) | **NAEST National Physics Test** (Cleared Prelims & Finals, IAPT / Prof. H.C. Verma) | **MTSO Olympiad** (Level 2 Qualifier).

Peer-Reviewed Publications & Proceedings

- **N. V. D. S. Sarat Chandra**, M. Nunna, Asha C. A., P. C. Sreekumar, **"Small Scale Raspberry Pi Based Autonomous Car: A Practical Guide to Understand Complexity."** in *Proc. IEEE ICRM*, 2025.
- **N. V. D. S. Sarat Chandra**, et al., **"Physics-Guided Diffusion with Thermodynamic Awareness for 3D Molecular Generation."** Submitted to *IEEE IES*, 2026.
- **N. V. D. S. Sarat Chandra**, et al., **"Domain-Aware Multi-Modal Industrial Diagnostics: Integrating Mechanical Heuristics with Conformal Uncertainty."** Submitted 2026.

Targeted Engineering Projects & Systems Architecture

Industrial GSM & 4G LTE Cellular IoT Gateway with Dual-Mode Failover — **Embedded Firmware Engineer**

Valetron A7670C 4G LTE Cat-1, ESP32, FreeRTOS, Circular DMA UART, 240V Relays

- **Core Impact:** <1.8s automatic network failover (WiFi <-> 4G LTE); opto-isolated 240V relay switching with RC snubbers.

Multi-Parameter Health Vitals Monitor & Real-Time IoT Data Streamer — **Biomedical Systems Developer**

MAX30102 PPG, ESP8266/ESP32, Fast-Mode I2C, 2nd-order IIR Bandpass, FreeRTOS

- **Core Impact:** <200ms real-time SpO2/HRV cloud telemetry; 9-clock I2C bus lockup recovery.

Pasupatha Autonomous Companion Robot Control Bus — **Head of Embedded Systems, Club Mirai**

ROS 2 Jazzy, STM32, ESP32, CAN 2.0B, C++ DLS Inverse Kinematics, HIL Testbed

- **Core Impact:** 50 Hz closed-loop control; <50us 5-DOF IK compute cycles (<0.4 deg repeatability).

Technical Skills & Core Competencies

Languages & Runtimes: C (C99/C11), C++ (C++17/20), Python (3.12), Bash, MATLAB, POSIX Sockets, FreeRTOS, Bare-Metal

Robotics, AI & Middleware: ROS 2 (Jazzy), CycloneDDS, Forward/Inverse Kinematics, OpenCV, PyTorch, PINNs, ONNX Runtime, Vitis AI EP

Hardware, Buses & Silicon: CAN Bus (2.0B), UART DMA, I2C (9-clock recovery), SPI, Valetron A7670C (4G LTE), STM32, ESP32, Raspberry Pi 5

Simulation & Tooling: ANSYS Fluent (CFD), SolidWorks, Fusion 360, Linux (Ubuntu Server), Docker, Git/GitHub, LaTeX